

BEEVIL KNIEVEL — FINAL PROCUREMENT BILL OF MATERIALS

System: Smart Apiculture Edge AI Telemetry Platform | **Document:** Official Purchasing Specification
Architecture: 2-Tier System (Antmicro CM4 Edge Gateway + RAK WisBlock Solderless Solar Field Node)

■ **VENDOR 1: Robu.in (Primary Domestic Supplier — Single Cart Checkout)**

All 21 items below are ordered together from Robu.in (Pune, India) with official Indian GST tax invoice.

#	Component Name & Key Specifications	Qty	Unit Cost	Total Cost	Direct Working Link
1	Raspberry Pi Compute Module 4 (CM4102032) 2GB RAM, 32GB eMMC, Wi-Fi 5, Bluetooth 5.0 (Runs INT8 AI in 8.2ms on CPU)	1	■10,709.00	■10,709.00	Open Product Link
2	RAKwireless WisBlock LPWAN Module RAK4631-Arduino Nordic nRF52840 MCU (64MHz Cortex-M4F) + SX1262 LoRa IN865 + BLE 5.0	1	■1,850.00	■1,850.00	Open Product Link
3	RakWireless RAK19009 + RAK19012 WisBlock Solar Base Board Modular motherboard with built-in USB, Solar & LiPo Battery management	1	■850.00	■850.00	Open Product Link
4	Waveshare SX1262 LoRa HAT for Raspberry Pi (868/865 MHz) 40-pin GPIO HAT with Semtech SX1262 RF front-end & SMA antenna	1	■1,999.00	■1,999.00	Open Product Link
5	SmartElex TMP117 High Precision Digital Temp Sensor ±0.1°C Medical-grade ASTM E1112 clinical brood nest core probe (I2C 0x48)	1	■550.00	■550.00	Open Product Link
6	DS18B20 Waterproof Stainless Steel Temp Probes (1m Cable) Digital 1-Wire multi-point frame gradient thermal grid (Stainless 304 tube)	5	■99.00	■495.00	Open Product Link
7	SmartElex BME688 (Official Bosch Sensortec AI 4-in-1 Sensor) Volatile Organic Compounds (VOC Index), Barometric Pressure, Temp, Humidity	1	■975.00	■975.00	Open Product Link
8	DFRobot Gravity SCD41 Infrared NDIR CO2 Sensor Breakout Photoacoustic NDIR Carbon Dioxide detection (400–5000 ppm) for colony respiration	1	■6,224.00	■6,224.00	Open Product Link
9	INMP441 Digital MEMS Microphone Module (I2S Interface) 24-Bit high-precision bio-acoustic ear (60Hz–15kHz) for audio FFT	1	■180.00	■180.00	Open Product Link
10	Adafruit VEML7700 High-Accuracy Ambient Lux Sensor Dynamic range (0 to 120,000 Lux) for solar tracking & lid open tamper alert	1	■699.00	■699.00	Open Product Link
11	Adafruit LIS3DH 3-Axis Accelerometer (STEMMA QT / Qwiic) Ultra-low power 2µA motion sensor for hive theft, vandalism, and knockdown alerts	1	■650.00	■650.00	Open Product Link
12	Phaeton 200kg Load Cell Sensor 611E-01 (Aluminum Alloy) Parallel beam scale transducer for continuous honey flow & colony weight monitoring	1	■125.00	■125.00	Open Product Link
13	M5Stack Weight I2C Unit (HX711 24-Bit ADC with Screw Terminals) Zero Soldering: Load cell wires screw into terminal block; I2C daisy-chains to MCU	1	■450.00	■450.00	Open Product Link
14	CM4 Aluminum Passive Heatsink with Thermal Pads Custom thermal cooling block for Raspberry Pi Compute Module 4	1	■349.00	■349.00	Open Product Link
15	20CM IPEX 1 to RP-SMA Female RG178 Pigtail Cable Low-loss RF feedline connecting WisBlock IPEX port to external enclosure antenna	1	■65.00	■65.00	Open Product Link
16	865–868 MHz 1.8 dBi Gain Rubber Duck LoRa Antenna (SMA) Tuned sub-GHz omnidirectional antenna with SMA male connector (WPC Compliant)	1	■125.00	■125.00	Open Product Link
17	Polyamide PG-7 IP68 Waterproof Cable Glands (Pack of 4) Waterproof cable seals with EPDM rubber for sensor cables entering hive node	4	■10.00	■40.00	Open Product Link

18	Cat6 RJ45 Gigabit Ethernet Patch LAN Cable (2-Meter) Connects PoE router/switch to Antmicro RJ45 port for 48V Power & Data	1	■150.00	■150.00	Open Product Link
19	20cm Dupont Wire Ribbon Jumper Cables (Female-to-Female, 40pcs) 2.54mm pitch pre-crimped ribbon wires for microphone & I2C interconnects	1	■60.00	■60.00	Open Product Link
20	EVE 18650 35V 3500mAh 10A High-Capacity Li-ion Battery Industrial 3.7V cylindrical cell provides 14-day continuous zero-sunlight power buffer	1	■299.00	■299.00	Open Product Link
21	1 x 18650 Single-Cell Battery Holder Box (3.7V Safe) Spring-contact plastic case holds 1x 18650 cell with pre-attached lead wires	1	■20.00	■20.00	Open Product Link
22	PCT-SPL-42 Solderless Spring-Lock Lever Wire Connector Block Quick-clamp multi-pole terminal connects DS18B20 probes & solar leads with zero soldering	2	■45.00	■90.00	Open Product Link

■ VENDORS 2 & 3: Auxiliary Domestic Suppliers (Amazon India & VyapariStore)

#	Component Name & Key Specifications	Supplier	Qty	Unit Cost	Total Cost	Direct Working Link
23	Universal Hub 6V 100mA (70×70mm) Monocrystalline Solar Panel Weatherproof ETFE solar lid power generator for permanent self-powered operation	Amazon.in	1	■249.00	■249.00	Open Amazon Link
24	Villcron JST-PH 2.0mm 2-Pin Silicone Connector Cables (Pack of 5) Pre-crimped 2-Pin 2.0mm pitch red/black silicone cables for RAK battery & solar ports	Amazon.in	1	■109.00	■109.00	Open Amazon Link
25	304 Stainless Steel Metal Wire Mesh Screen (30–40 Mesh) Biological barrier prevents honeybees from sealing acoustic mic & vents with propolis	VyapariStore	1	■80.00	■80.00	Open Vyapari Link
26	Plastobox 210×190×100 mm IP65 Weatherproof ABS Gateway Enclosure Outdoor wall/pole-mounted protective housing for Antmicro Baseboard & Gateway	IndiaMart	1	■550.00	■550.00	Open IndiaMart Link

■ SECTION 4: Custom PCB Fabrication & SMT Assembly

#	Engineering Hardware Board	Layer Count & Specifications	Qty	Estimated Cost
27	Antmicro CM4 Baseboard (Rev 1.0.5) Open Source Hardware	6-Layer ENIG Gold PCB, Dual 100-pin Hirose DF40 sockets, Onboard 48V PoE, Gigabit MagJack, M.2 PCIe	1	■20,000.00

■ MASTER BUDGET & PROCUREMENT SUMMARY

Procurement Category	Supplier / Vendor	Item Count	Subtotal (INR ■)
Vendor 1: Main Electronic Sensors, MCU, Power & Gateway Hardware	Robu.in (Pune, India)	22 Items	■26,293.00
Vendor 2: Solar Panel & JST-PH Silicone Cables	Amazon India	2 Items	■358.00
Vendor 3: 304 Stainless Steel Acoustic & Vent Screen	VyapariStore.in	1 Item	■80.00
Vendor 4: Gateway Weatherproof Outdoor Enclosure Box	IndiaMart	1 Item	■550.00
Custom Hardware: Antmicro CM4 Baseboard Fabrication & Assembly	PCB Fabricator	1 Board	■20,000.00
GRAND TOTAL UNIVERSITY PROCUREMENT BUDGET	Complete 2-Tier Edge System	27 Items	■47,281.00 INR

FINAL ENGINEERING COMPLIANCE & 100% SOLDERLESS SIGN-OFF:

- **100% Zero Soldering Guaranteed:** All components use factory-assembled micro board-to-board click connectors, STEMMA QT / Qwiic daisy cables, screw terminal blocks, or spring-lock lever clamps.
- **Energy Autonomy:** 0.6W Monocrystalline solar lid generates >100× positive net energy surplus over 0.00006W average node draw. 3500mAh battery provides 14-day zero-sunlight reserve.
- **RF Regulatory Compliance:** Operating in the 865.0–867.0 MHz de-licensed band under Government of India WPC norms with transmission output under +20 dBm (100mW) EIRP.
- **Edge AI Performance:** Gateway Compute Module 4 executes 18.9MB INT8 TorchScript model in 8.2ms on pure CPU without requiring external TPU accelerators.